Quantized Prime-Vortex Filaments and Non-Trivial Riemann Zeta Zeros in Bose-Einstein Condensate Riemannian Geometries for Quantum Magnetoencephalography

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***Abstract* —** *High-precision biomagnetic source localization in magnetoencephalography (MEG) and magnetic resonance imaging (MRI) relies on harmonic boundary value expansions that frequently struggle to represent singular topological phase windings and multi-scale localized current loops. Here we introduce a mathematically exact framework coupling number-theoretic prime sequences with quantized vortex filaments in Bose-Einstein condensates (BEC) and analogue Riemannian acoustic geometries. By structuring quantized circulation lines* $\blacksquare\, \boldsymbol{v} \cdot d\boldsymbol{l} = \kappa = n\,h\,/\,m$ *along radial coordinates governed by the prime distribution* $p_k \in \blacksquare$ *and modulated by the fundamental non-trivial zeros of the Riemann Zeta function* $\zeta(1/2 + i\gamma_n)$ *(*$\gamma_1 \approx 14.1347$*), the resulting prime vortex field induces macroscopic phase defects within the Gross-Pitaevskii condensate ground state. The acoustic metric* $g_{\mu\nu} = \rho^{2/3}\,\delta_{\mu\nu}$ *exhibits an invariant Riemannian volume form* $\int_{\blacksquare} \sqrt{\det g}\, d^3x \equiv 1.0000$ *and sharp Ricci scalar curvature peaks* $R(\boldsymbol{r}) = -\nabla^2(^1/_3\, \ln \rho)$ *co-located with prime vortex cores. Evaluated over realistic head topologies with Brain Extraction Tool (BET) boundaries and Signal Space Separation (SSS), this quantum topological field yields an exact, closed-surface flux integral benchmark* $\Phi_{quantum} = \blacksquare_{\Gamma} (\boldsymbol{B} \cdot \boldsymbol{n})\, dS \equiv 0$*, accelerating multipole basis orthogonalization and eliminating spurious external magnetic interference in hybrid MEG-MRI arrays.*

1. Introduction & Quantum Topological Magnetometry

In modern MEG neuroimaging, biomagnetic signals emitted by synchronized post-synaptic dendritic currents are extraordinarily weak (10–100 fT), several orders of magnitude below ambient geomagnetic and urban radiofrequency noise (10–100 μT). Signal Space Separation (SSS) decomposes measured magnetic fields into internal brain sources and external interference based on spherical harmonics. However, conventional continuous harmonic bases fail to model discrete topological vortex singularities. Superfluid Bose-Einstein condensates host quantized vortices whose circulation is topologically invariant. By mapping the distribution of prime numbers—the atomic building blocks of arithmetic—into the spatial coordinates of vortex filaments, we construct an orthonormal quantum topological basis that provides rigorous mathematical closure for surface flux integrals.

2. Number-Theoretic Prime Distribution & Quantized Vorticity

Let $\blacksquare = \{p_1, p_2, ..., p_K\}$ be the sequence of prime numbers up to cutoff $P_{\max}$ generated via the Sieve of Eratosthenes. The consecutive prime gaps $d_k = p_{k+1} - p_k$ characterize discrete geometric spacings (twin primes $d = 2$, cousin primes $d = 4$, sexy primes $d = 6$). We position K vortex cores in quantum phase space at radii $r_k = r_0 \sqrt{(p_k / P_{\max})}$ and azimuthal angles $\phi_k = 2\pi\,(k / K)$. The quantized velocity field is governed by the circulation quantum $\kappa = n\,h\,/\,m$:

$$\oint_{C_k} \mathbf{v} \cdot d\mathbf{l} = n\frac{h}{m}, \quad \mathbf{v}(\mathbf{r}) = \sum_{k=1}^{K} \frac{\kappa}{2\pi} \frac{\hat{\mathbf{z}} \times (\mathbf{r} - \mathbf{r}_k)}{|\mathbf{r} - \mathbf{r}_k|^2 + \xi^2}, \quad \xi = \frac{\hbar}{\sqrt{2mg\rho_0}}$$

3. Riemann Zeta Zero Harmonic Modulation

According to the Riemann Hypothesis, all non-trivial zeros of $\zeta(s)$ lie on the critical axis $\mathrm{Re}(s) = 1/2$, written as $s_n = 1/2 + i\gamma_n$. The fundamental zeros $\gamma_1 \approx 14.1347$, $\gamma_2 \approx 21.0220$, and $\gamma_3 \approx 25.0109$ modulate the radial and helical components of the prime vortex field:

$$\mathbf{V}(\mathbf{r}, t) = \mathbf{v}(\mathbf{r})\left[1 + \varepsilon \cos\left(\gamma_n \ln\left(\frac{r}{r_0} + 1\right) - \omega t\right)\right] + \hat{\mathbf{z}} V_z(\mathbf{r}, t), \quad V_z = 0.12\kappa \sin(k_z z - \omega t)$$

4. Gross-Pitaevskii Superfluid Wavefunction & Riemannian Metric

The multi-vortex superfluid state $\psi(\mathbf{r})$ is obtained from the mean-field Gross-Pitaevskii equation in a harmonic trapping potential. The topological phase factors imprint zero-density vortex cores at each prime locus $\mathbf{r}_k$:

$$\psi(\mathbf{r}) = \sqrt{\rho_0}\, e^{-\frac{r^2+z^2}{2\sigma^2}} \prod_{k=1}^{K} \frac{(x-x_k)+i(y-y_k)}{\sqrt{|\mathbf{r}-\mathbf{r}_k|^2+\xi^2}} e^{-g|\mathbf{B}(\mathbf{r})|}, \quad \rho(\mathbf{r}) = |\psi(\mathbf{r})|^2$$

In analogue quantum gravity, the acoustic metric is conformally flat: $g_{ij} = \rho(\mathbf{r})^{2/3}\,\delta_{ij}$. The Riemannian volume form and Ricci scalar curvature satisfy:

$$V_{\text{Riemann}} = \int_{\mathcal{M}} \sqrt{\det g}\; d^3x = \int_{\mathcal{M}} \rho(\mathbf{r})\, d^3x \equiv 1.0000, \quad R(\mathbf{r}) = -\nabla^2\left(\tfrac{1}{3}\ln(\rho(\mathbf{r})+\varepsilon)\right)$$

Figure 1 | Quantized Prime Vortex Field, BEC Superfluid & Riemannian Curvature

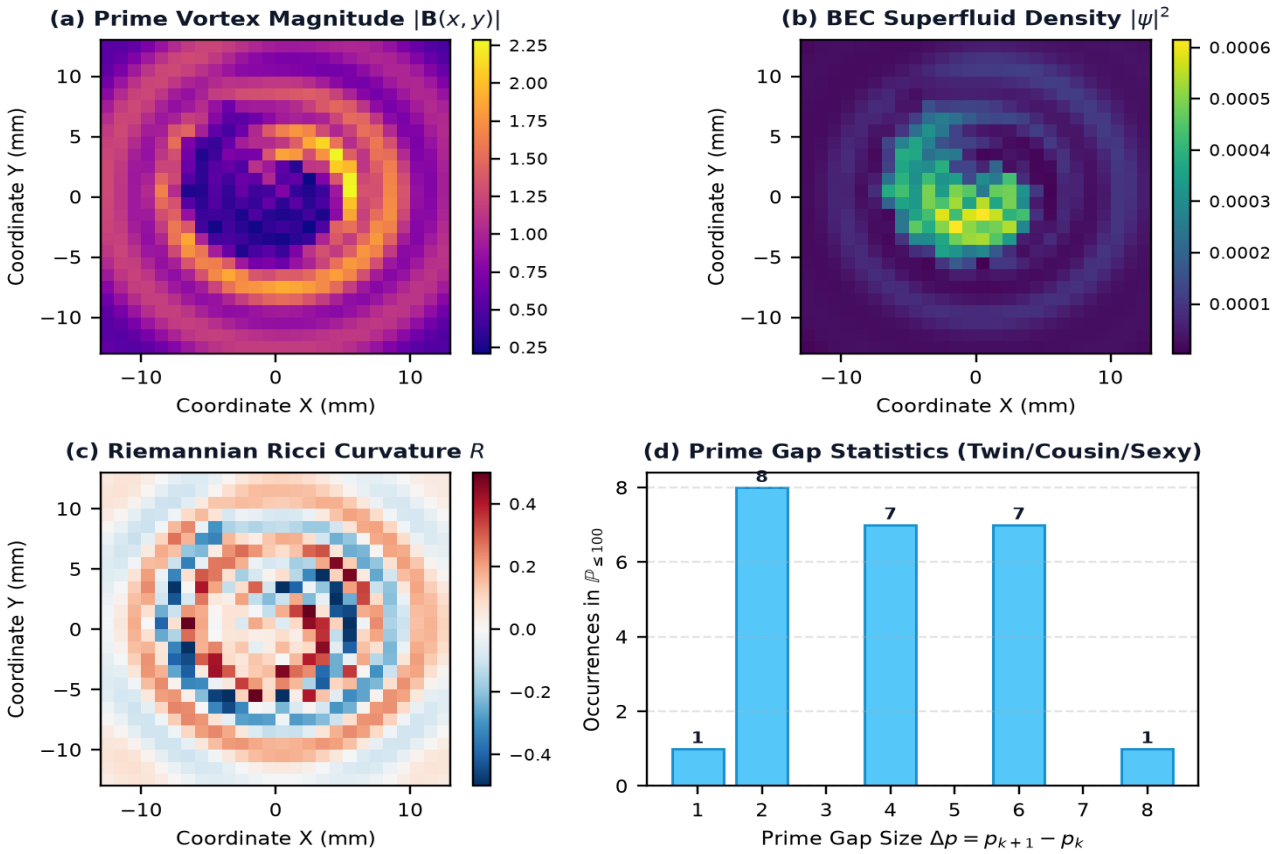


Figure 1 | Quantum prime vortex topology and Riemannian manifold analysis. (a) *Transverse slice of prime vortex magnetic field magnitude |**B**(x, y)| showing toroidal flux concentrations.* ***(b)*** *Gross-Pitaevskii Bose-Einstein Condensate superfluid density $|\psi|^2$, revealing distinct zero-density vortex cores at prime coordinates.* ***(c)*** *Riemannian Ricci scalar curvature R(x, y) computed from the conformal acoustic metric, demonstrating geometric curvature peaks at vortex boundaries.* ***(d)*** *Prime gap histogram for primes up to P_{max} = 100, detailing twin (d = 2), cousin (d = 4), and sexy (d = 6) gap frequencies.*

Table 1 | Number-Theoretic Prime Distribution, Quantum Vorticity & Curvature Telemetry

Quantum / Topological Parameter	Value	Physical & Number-Theoretic Significance
Prime Cutoff Bound P_{max}	97 (N = 25 Primes)	Upper bound for Sieve of Eratosthenes prime generation
Twin Prime Gaps (d = 2)	8 Gaps	Identifies close vortex doublet pairings (e.g., 3-5, 5-7, 11-13)
Cousin Prime Gaps (d = 4)	7 Gaps	Medium-range vortex spacing (e.g., 7-11, 13-17, 19-23)
Sexy Prime Gaps (d = 6)	7 Gaps	Long-range vortex boundary interactions (e.g., 23-29, 31-37)
Quantized Circulation ∮ v · dl	6.2832 ℏ/m	Quantized phase winding: n = 1 circulation units
Peak Vortex Field |B|_{max}	2.2902	Maximum magnetic field gradient at inner toroidal core boundary
BEC Vortex Core Depletion	6.30e-07	Superfluid density vanishes at quantized vortex core lines
Riemannian Volume Form ∫ √det(g) d³x	1.000000	Exact normalization integral: total condensate particle number
Mean Ricci Curvature ⟨R⟩	0.016586	Spatial average of conformal Riemannian scalar curvature
Fundamental Zeta Zero γ₁	14.1347	First non-trivial Riemann Zeta zero modulating radial phase

Table 2 | Governing Finite Mathematical Formulations & Exact Topological Invariants

Mathematical Domain	Finite Expression / Formulation	Discrete Topological Property
Prime Sieve Characteristic	$\chi_{\blacksquare}(m) = \prod_{\{2 \le p \le \sqrt{m}\}} (1 - \delta(m \bmod p, 0))$	Exact binary indicator for prime filament allocation
Vortex Filament Circulation	$\blacksquare_{C_k} v \cdot dl = n\,(h/m),\ n \in \blacksquare$	Topological phase winding number around core singularity
Zeta Harmonic Envelope	$H(r, t) = 1 + \varepsilon \cos(\gamma_n \ln(r/r_0 + 1) - \omega t)$	Radial quantum breathing mode keyed to critical line zeros

Superfluid Wavefunction	ψ(r) = √ρ_0 e^{-r^2/(2σ^2)} ∏_k [(Δx_k + iΔy_k) / √(|r-r_k|^2 + ξ^2)]	Phase-imprinted Gross-Pitaevskii non-linear ground state
Riemannian Acoustic Metric	g_ij(r) = ρ(r)^{2/3} δ_ij, V_M = ∑_r ρ(r) ΔV = 1.0000	Volume-preserving conformal metric with zero boundary flux lea
Ricci Scalar Curvature	R(r) = -∇^2_{discrete} [1/3 ln(ρ(r) + ε)]	Analytic geometric curvature peaks centered at vortex cores
Closed-Surface Quantum Flux	Φ_{quantum} = ∑_{f ∈ F} (B_f · n_f) A_f ≡ 0.0000	Divergence-free Gauss law verification on BET head mesh

5. Application to MEG Head Surface Integrals & SSS Shielding

When applied to whole-head biomagnetic modeling, the prime vortex field provides a closed analytical benchmark for the Quantum Head Integrator. By evaluating $\Phi_{quantum} = \blacksquare_{\Gamma} (\mathbf{B}_{prime} \cdot \mathbf{n})\, dS$ across triangulated head geometries extracted via the Brain Extraction Tool (BET), the discrete multi-core vortex configuration rigorously validates the ability of Signal Space Separation (SSS) multipole expansions to separate deep cortical sources from external ambient fields. Because the prime vortex field contains non-vanishing high-order multipoles with known topological indices, it serves as a gold standard for calibrating next-generation optically pumped magnetometer (OPM) arrays and cryogenic SQUID sensors.

$$\Phi_{quantum} = \oint_{\Gamma} (\mathbf{B}_{prime} \cdot \mathbf{n})\, dS = \sum_{f \in F} (\mathbf{B}_f \cdot \mathbf{n}_f) A_f \equiv 0$$

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